e-Hospital AI-Twin: Showcase Video Presentation Script

Closed-Loop Healthcare Platform: Wearable Telemetry, WhatsApp Care & Agentic AI

Innovator & Lead Developer: Suganthan Subramanian | **Affiliation:** Faculty of Engineering, University of Ottawa
Project: e-Hospital AI-Twin | **Target:** Healthcare Innovation & Barrier Showcase Competition

Project Mission: Solving the healthcare crisis at both ends: eliminating clinician administrative burnout ('documentation tax' of 2 hours paperwork per 1 hour of patient care) while overcoming the remote patient monitoring barrier (wearable data trapped in silos; 70%+ portal abandonment). This system closes the loop via a cross-platform Flutter app, background Apple HealthKit sync, zero-barrier WhatsApp messaging, and a multi-agent clinical reasoning backend hosted on AWS App Runner.

Complete App Capability Map (100% Feature Coverage)

Component / Module	Key Capabilities Covered	Clinical Value
1. Patient Dashboard	Real-time telemetry sync, connection status, vitals quick-cards, direct WhatsApp care launcher.	Zero friction entry point for daily wellness overview.
2. Wearable & HealthKit Sync	Background ingestion of resting HR, blood pressure, SpO2, and glucose from Apple Watch / Wear OS.	Continuous monitoring; ends manual logging and eliminates data silos.
3. Sleep & Stress Engine	Calculates autonomic recovery and stress scores from sleep stages and resting HRV.	Early marker for burnout, chronic inflammation, and cardiovascular risk.
4. AI Health Chat	Conversational inquiry against live AWS MySQL records; suggests prompt chips.	Patient health literacy and self-management without clinical visits.
5. Lab Report Interpreter	Extracts biomarkers from blood work/PDFs (HbA1c, lipid panel, creatinine); flags abnormal values.	Translates intimidating lab jargon into clear, actionable explanations.
6. Nutrition & Meal Monitor	Logs meals; calculates glycemic/sodium impact scored against hypertension & diabetes guidelines.	Proactive chronic disease dietary adherence.
7. WhatsApp Care Channel	Meta Cloud Webhook parsing biometrics from text/voice notes directly into AWS database.	Universal accessibility for elderly/rural patients; 0% portal drop-off.
8. Urgent Care Triage Queue	Evaluates inbound telemetry against Canadian guidelines to flag dangerous anomalies.	Prioritizes deteriorating patients for immediate clinical intervention.
9. Ambient Voice Scribe	Passively listens to doctor-patient dialogues; generates SOAP notes & OHIP billing codes.	Saves 1.5–2 hours of physician charting daily; stops burnout.
10. 3D Patient Digital Twin	Maps physiological parameters onto interactive 3D anatomical organ meshes (Three.js/WebGL).	Empowers shared clinical decision-making through intuitive 3D visuals.
11. Privacy & Cloud Core	FastAPI on AWS App Runner with local on-premise Ollama (Gemma 2 / Llama 3) support.	Strict PHIPA/PIPEDA compliance; zero PHI cloud leakage.

Complete 4:45 Minute Video Speech Script

Word-for-word spoken transcript with exact visual cues and on-screen actions:

0:00 – 0:40 (40s) — Introduction & The Healthcare Barrier

Visual Action: *Webcam visible on camera or opening title slide*

"Hello everyone. My name is Suganthan Subramanian, and I am a graduate researcher in Biomedical Engineering and eHealth at the University of Ottawa. Today, I am proud to present e-Hospital AI-Twin—a comprehensive digital health platform designed to solve two massive healthcare bottlenecks:

1. Physician Burnout: Doctors spend nearly two hours on EHR paperwork for every single hour of patient care.
2. Fragmented Patient Monitoring: Patients collect vital signs on Apple Watches and wearables, but that data never reaches their hospital charts. Meanwhile, existing hospital portals suffer from over 70% patient drop-off because they are too complicated for everyday patients.

To solve this, I engineered a closed-loop ecosystem connecting native mobile apps, ubiquitous messaging, cloud EHRs, and clinical AI."

0:40 – 1:30 (50s) — Screen 1: Flutter Dashboard & Wearable Sync

Visual Action: *Screen recording showing Flutter Patient Dashboard and live sync status*

"Let's look at our Flutter mobile application. Right on the Patient Dashboard, you see our real-time synchronization engine. Through native Apple HealthKit and Wear OS integrations, the app continuously collects resting heart rate, blood pressure, blood oxygen, and blood glucose in the background—requiring zero manual typing from the patient.

It also runs our Sleep and Stress Analytics Engine, combining resting heart rate variability and sleep duration to generate a personalized autonomic recovery score. All of this telemetry streams live into our centralized AWS App Runner cloud database."

1:30 – 2:40 (70s) — Screen 2: Multi-Module AI Health Assistant

Visual Action: *Screen recording clicking through AI module picker: Chat, Report Interpreter, Nutrition*

"Next, let's open the AI Health Assistant. You'll notice three dedicated clinical modules:

Module 1: The Interactive Health Chat: Patients can conversationally query their physiological history. For example, clicking 'What is my latest heart rate?' causes our LangGraph multi-agent engine to query our central AWS MySQL tables, check the numbers against clinical guidelines, and return empathetic, accurate health guidance.

Module 2: The Lab Report Interpreter: When patients receive complex blood work or diagnostic PDFs, this module extracts biomarker values (such as HbA1c, lipid panels, and creatinine), flags out-of-range metrics, and translates intimidating medical jargon into clear, reassuring explanations.

Module 3: The Nutrition & Chronic Disease Monitor: Patients can log meals or dietary habits. The AI scores their nutritional intake directly against their specific chronic conditions—for example, alerting a hypertensive patient about sodium spikes, or evaluating glycemic index for a diabetic patient."

2:40 – 3:30 (50s) — Screen 3: WhatsApp Channel & Urgent Care

Visual Action: *Screen recording showing 1-Tap WhatsApp button and WhatsApp Web live conversation*

"Now, what about elderly or non-technical patients who don't use mobile apps? This is our biggest accessibility breakthrough: The WhatsApp Healthcare Channel.

Right from the Flutter dashboard, there is a 1-tap Care Team escalation button that opens a direct line to clinicians.

Furthermore, patients can simply open WhatsApp and text: 'My BP is 135/88 and my heart rate is 74'. Our FastAPI webhook verifies Meta's security signatures, extracts the numerical biometrics via regex and clinical heuristics, and commits them straight into the

hospital's central EHR.

If a vital sign exceeds critical safety limits, it automatically feeds into our Urgent Care Triage Queue, prioritizing high-acuity patients at the top of the clinician's review list."

3:30 – 4:15 (45s) — Screen 4: Ambient Voice Scribe, 3D Digital Twin & Privacy

Visual Action: *Screen recording showing FastAPI Swagger docs, backend code, or 3D viewer*

"For the healthcare provider, the platform eliminates documentation fatigue through our Ambient Clinical Voice Scribe. It passively listens to doctor-patient consultations, generates structured SOAP notes, and suggests provincial billing codes (such as OHIP) in seconds.

To support shared decision-making, telemetry also maps onto an interactive 3D Patient Digital Twin, giving doctors and patients an intuitive visual representation of organ health.

Crucially, our architecture is built around Privacy-by-Design: it supports running fully on-premise using local LLMs through Ollama (hosting Gemma 2 and Llama 3 models), ensuring strict compliance with Canadian health privacy laws like PHIPA and PIPEDA."

4:15 – 4:45 (30s) — Conclusion & Summary

Visual Action: *Switch back to webcam or final slide with contact and roadmap*

"In summary, e-Hospital AI-Twin is not just a concept—it is a functional, end-to-end ecosystem: continuous biometrics from Apple HealthKit, zero-barrier messaging on WhatsApp, AI-driven lab interpretation and nutrition monitoring, and ambient clinical scribing that saves doctors two hours every single day—all unified on high-availability AWS cloud infrastructure.

As a graduate engineering researcher at the University of Ottawa, I engineered this entire platform to prove that modern AI and accessible consumer tools can work hand-in-hand to restore humanity to healthcare.

Thank you very much!"

Pre-Recording Quick Checklist

Step	Preparation Item
1. Windows	Open Flutter app in Chrome, WhatsApp Web, and Swagger docs (localhost:8000/docs).
2. Tool	Use Loom (free, records screen + camera bubble) or Zoom ('Record to Computer').
3. Audio	Use headset/USB mic in a quiet room. Speak at an energetic, conversational pace.
4. Time	Target 4:30 to 4:45 minutes. Maximum allowed is 5 minutes (300 seconds).
5. Export	Export as MP4 (under 1GB) and upload directly to the Google Form.